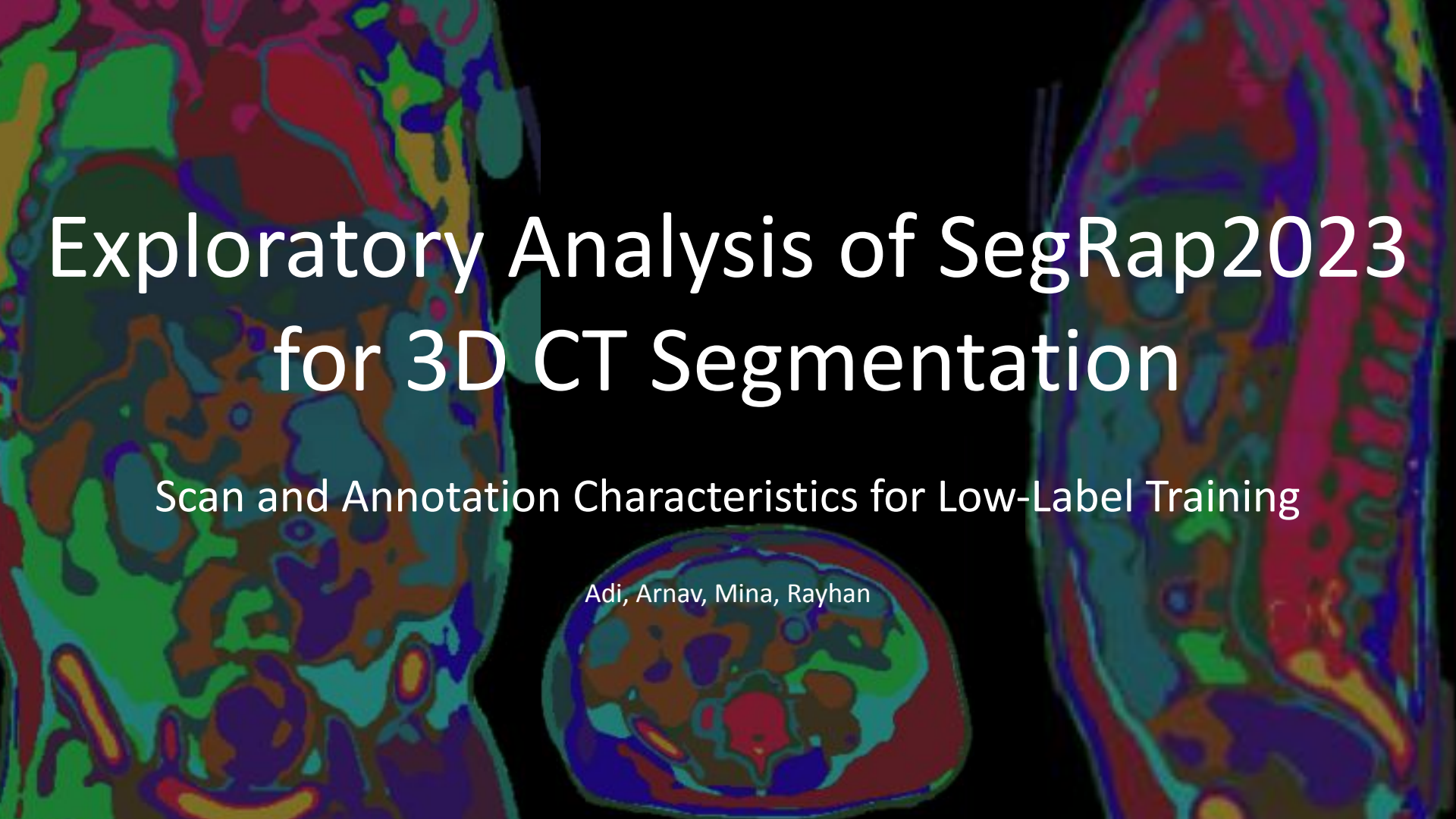




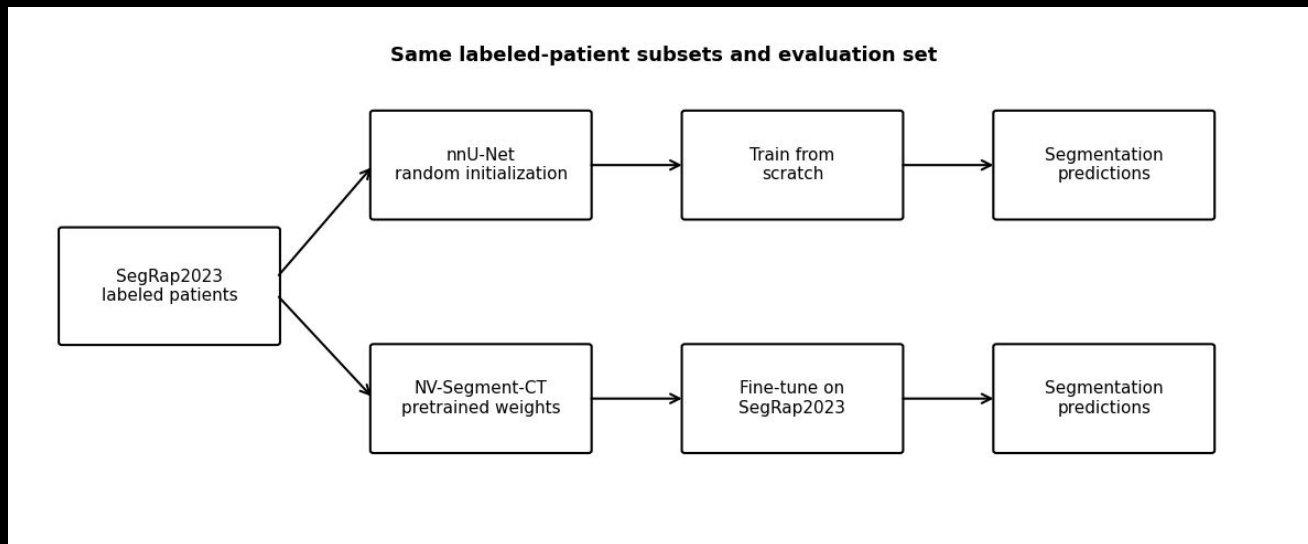
Exploratory Analysis of SegRap2023 for 3D CT Segmentation

Scan and Annotation Characteristics for Low-Label Training

Adi, Arnav, Mina, Rayhan

Project Recap

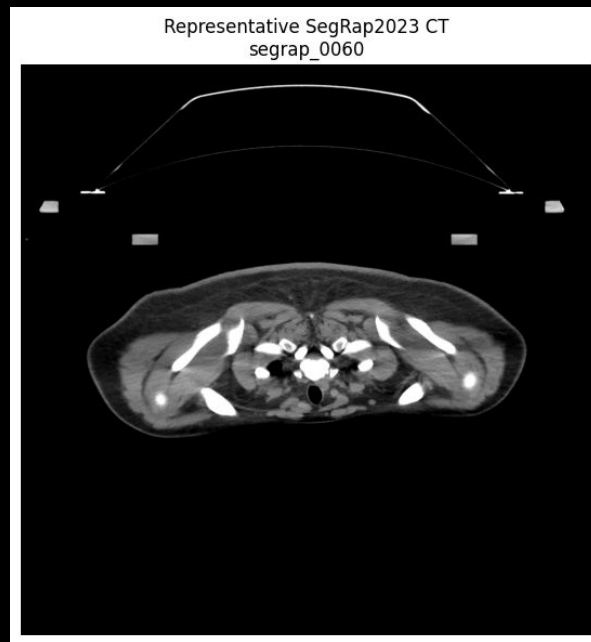
- **Research Question:** Does pretraining improve 3D CT segmentation performance when labeled training data are limited?
- **nnU-Net** (train from scratch) vs. **NV-Segment-CT** (fine-tune foundation model)



SegRap2023 Dataset Overview

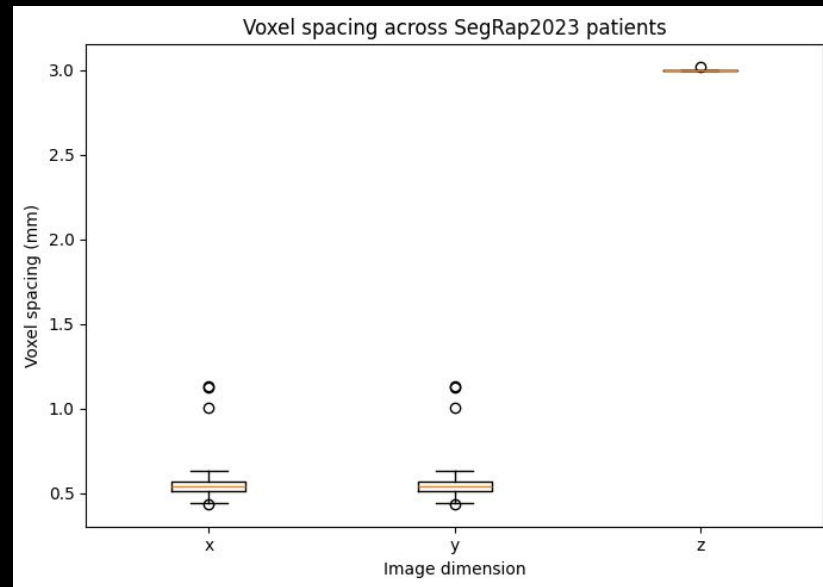
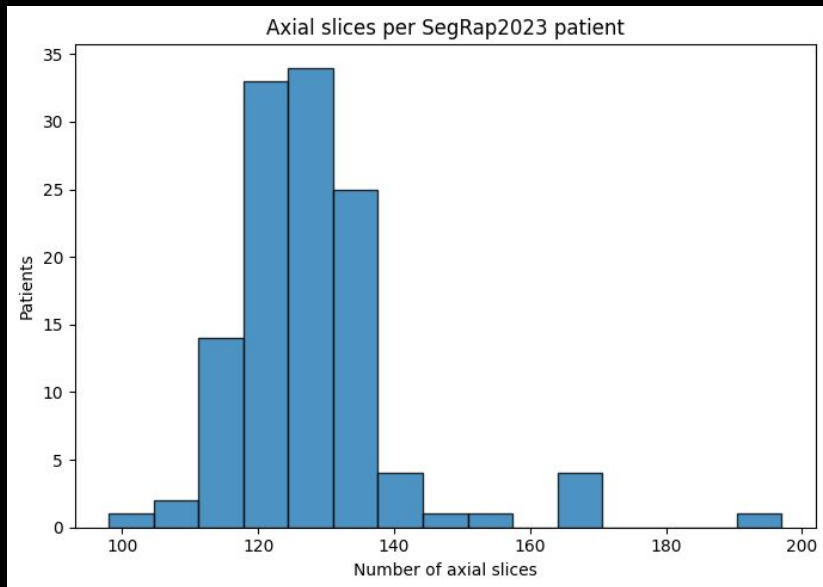
Why SegRap2023? Its patient-level 3D CT scans and detailed anatomical annotations allow us to systematically reduce the amount of labeled training data while keeping evaluation consistent

- 120 + 20 + 60 patients (training, validation, test)
- Head & neck CT
- ncCT + ceCT
- 3D patient-level volumes
- Detailed anatomical labels



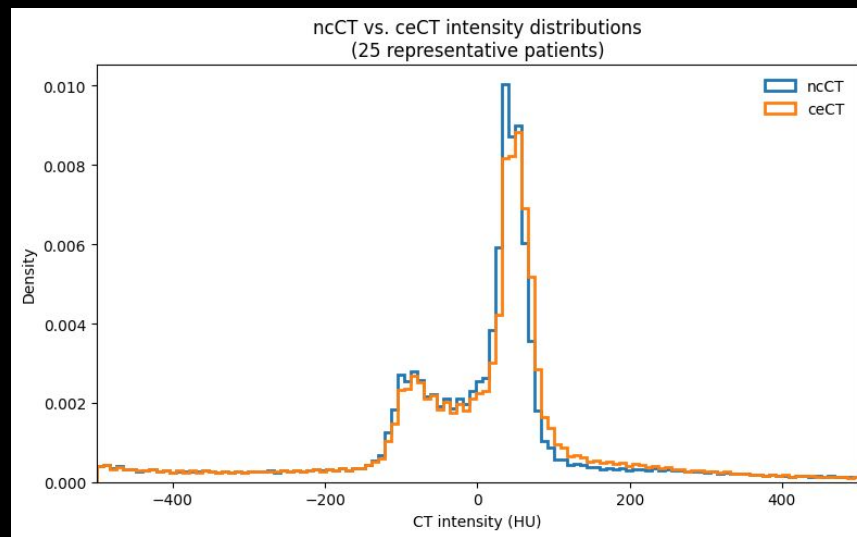
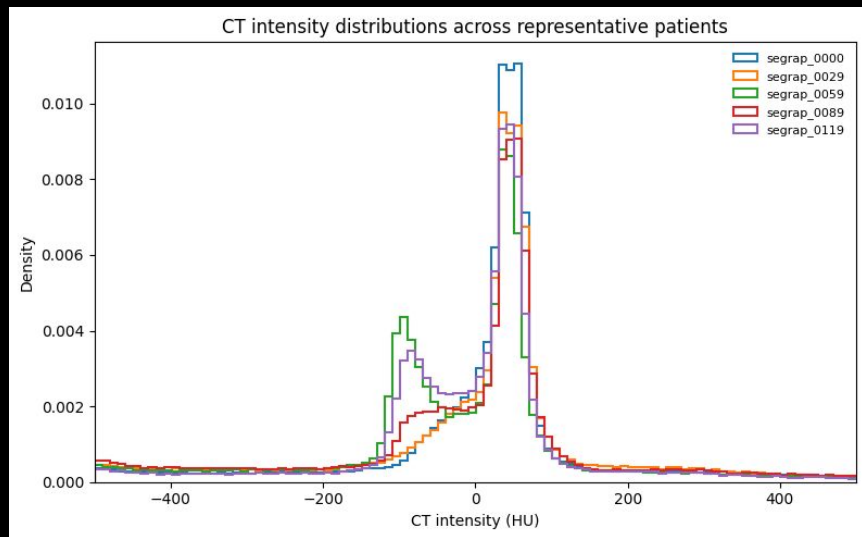
SegRap2023 Slices Vary Spatially

Takeaway: Variation in scan dimensions and spatial resolution determines whether spatial standardization is needed before model training.



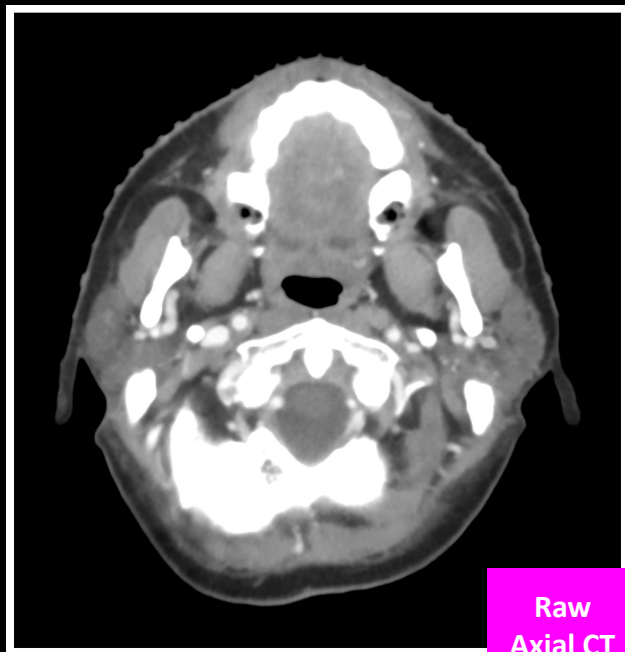
CT Intensity Distributions and Normalization

Takeaway: Comparing intensity distributions across scans and modalities will inform the intensity preprocessing used before training.

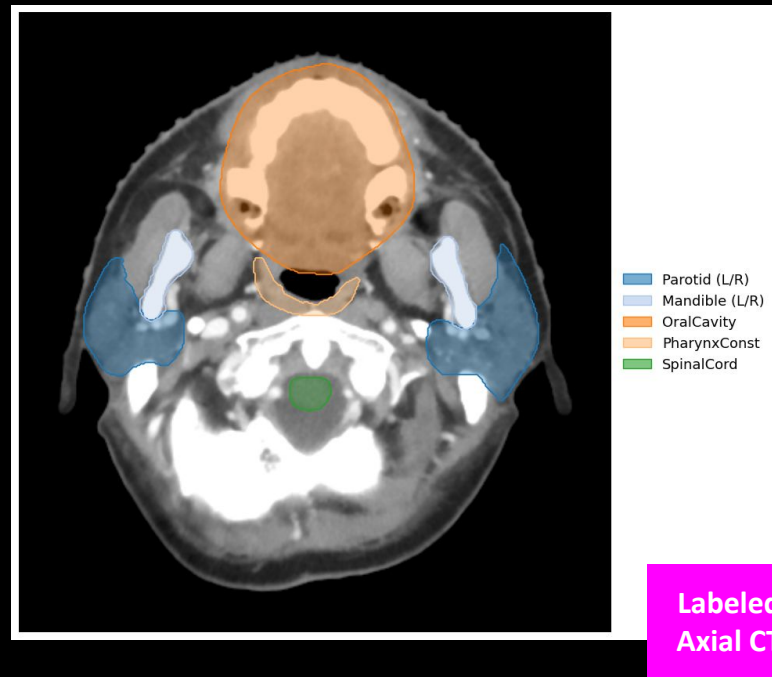


Visualizing SegRap2023 Annotations

Takeaway: SegRap2023 includes labeled structures across individual 2D slices

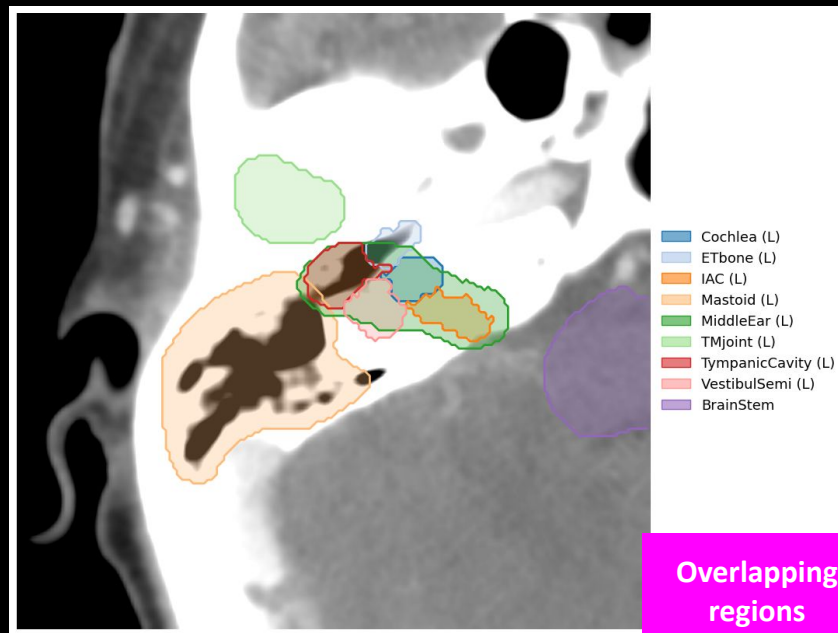
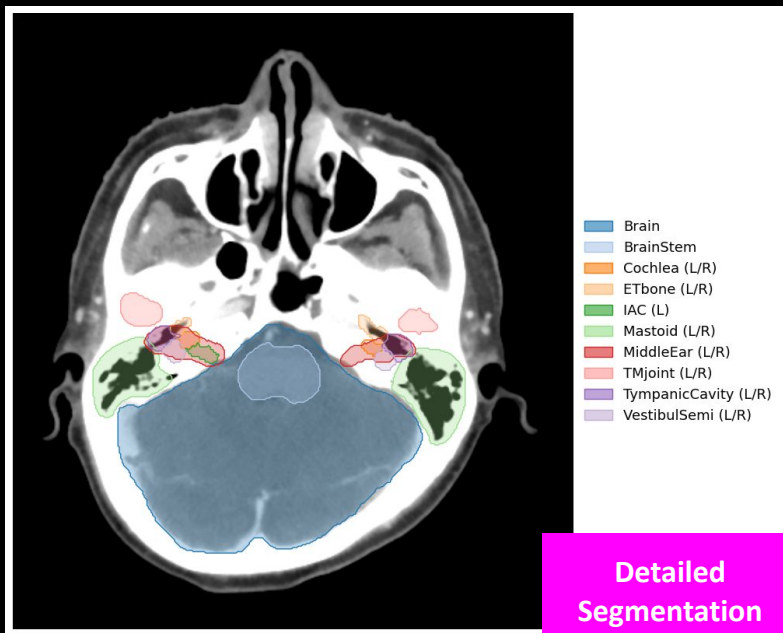


Individually
annotated
regions



SegRap2023 Annotations Include Complexity

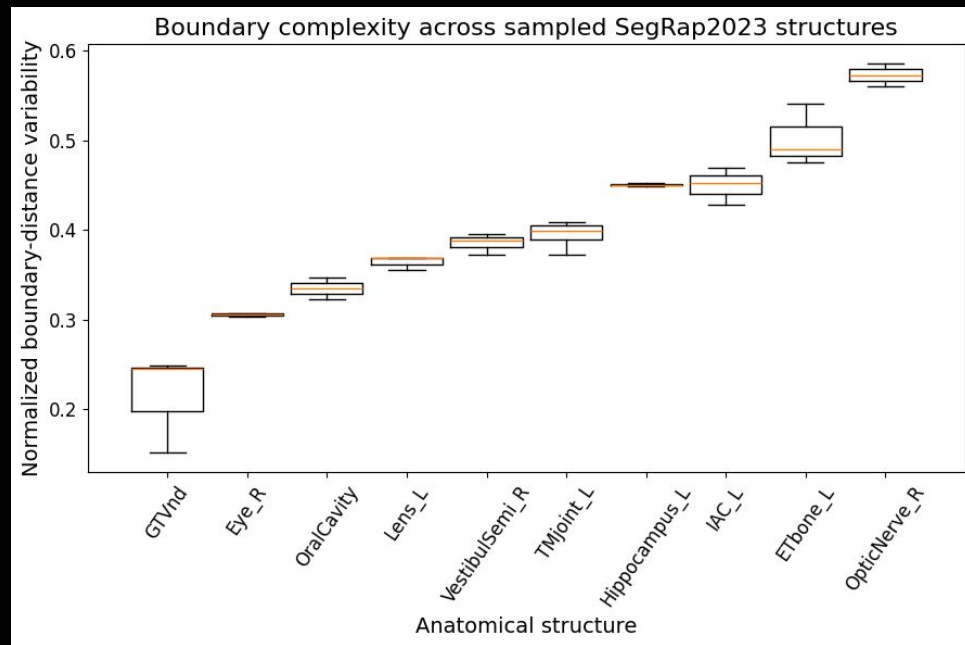
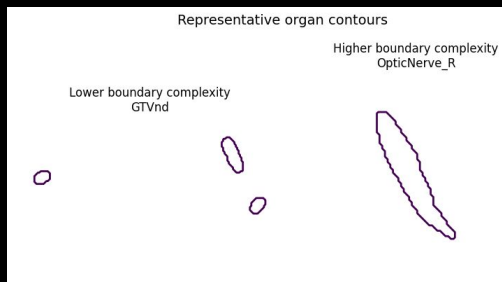
Takeaway: SegRap2023 contains structures with substantially different spatial relationships, including nested and overlapping anatomical regions.



Annotation Boundaries Vary Between Organs

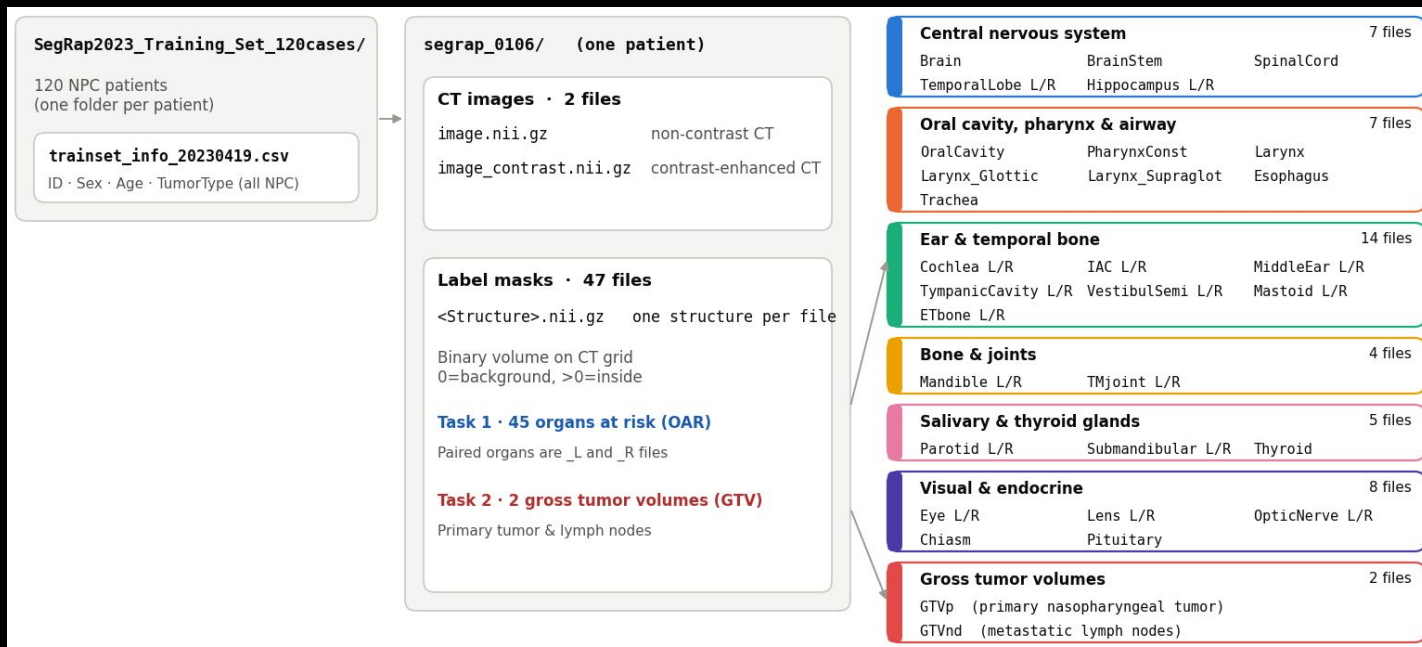
Takeaway: Differences in boundary geometry provide an additional dimension of segmentation difficulty beyond structure size alone.

Evaluation implication: motivates complementing overlap-based metrics with a boundary-sensitive metric.



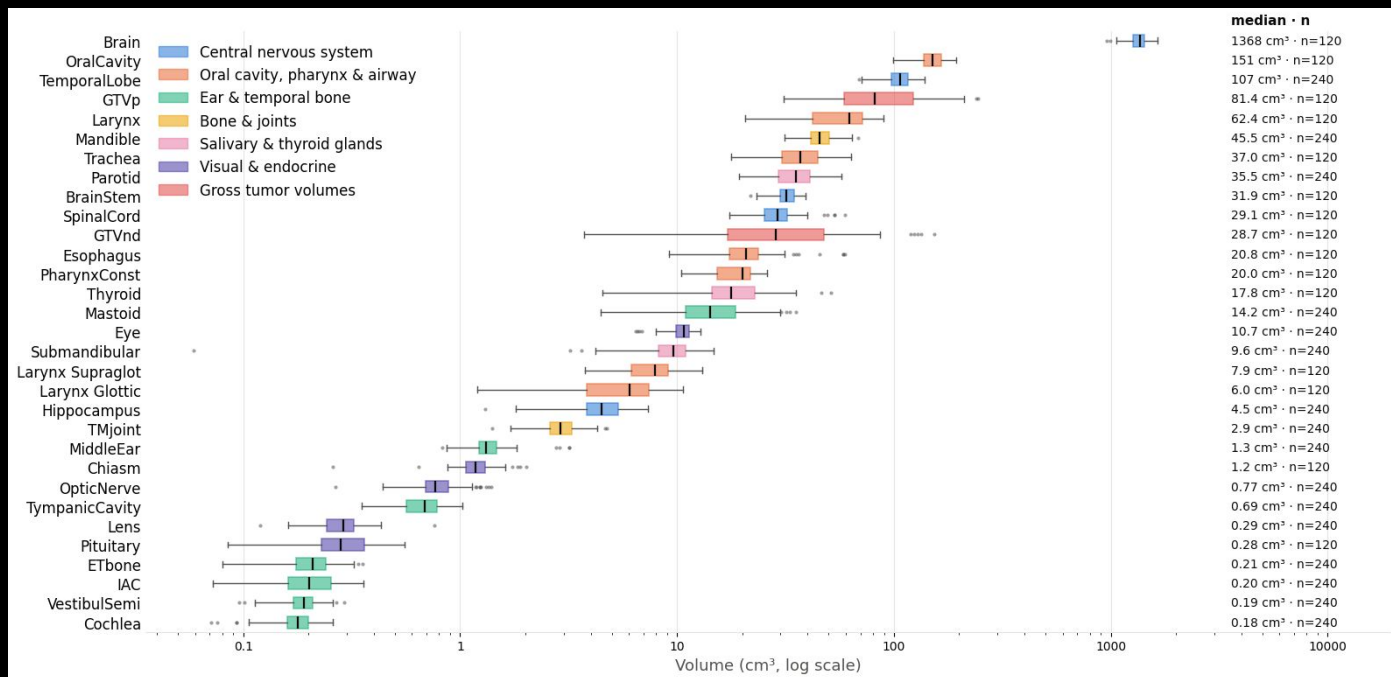
Annotation Characteristics

Takeaway: SegRap includes annotations on many different head neck and neck regions with metadata



SegRap Labeled Regions Have Size Differences

Takeaway: Large differences in structure size may cause segmentation performance to vary across anatomical structures.



NV-Segment-CT validation of training labels

Takeaway: NV-Segment-CT allows for automatic segmentation of 127 classes, but only a subset may be applicable for predicting SegRap2023 labels

Dataset ID	Type	# cases used
TCIA Pancreas CT [47]	Abdominal CT organs	80
AbdomenCT-1K [31]	Abdominal CT organs	1,050
AMOS22 [23]	Abdominal CT organs	300
MSD Task 3,6,7,8,9,10 [3]	Various lesions	945
CT-ORG [46]	Lung, bones, liver, kidneys, bladder	136
TotalSegmentator [12]	Many anatomic structures	1,228
CRML-CT [52]	Liver, colorectal liver metastases	197
VerSe [50]	Vertebral labelling	374
AeroPath [53]	Airways and lungs	27
Bone lesion (in-house)	bones	296
LIDC-IDRI [4]	Unannotated, lung cancer screening thoracic	470
COVID-19 [16]	Unannotated, chest	524
TCIA Colonography [25]	Unannotated, abdomen	1,440
StonyBrook COVID19 CT [49]	Unannotated, chest	1,274
NLST [56]	Unannotated, chest	3,113

Table 1. Summary of datasets used for model training.

SegRap2023 labels vs. NV-Segment-CT (VISTA3D) supported classes

Share of the 47 SegRap2023 labels: 11% / 21% / 38% / 30%

Direct match (class exists in VISTA3D) (5)

Brain → #22 Esophagus → #11 Spinal cord → #121 Thyroid → #126 Trachea → #57

Sub-region of, or continuous with, a supported organ (10)

Brainstem Hippocampus L Hippocampus R Larynx Larynx glottic Larynx supraglottic Oral cavity Pharynx Temporal lobe L
Temporal lobe R

Skull base / temporal bone (nearest class: skull) (18)

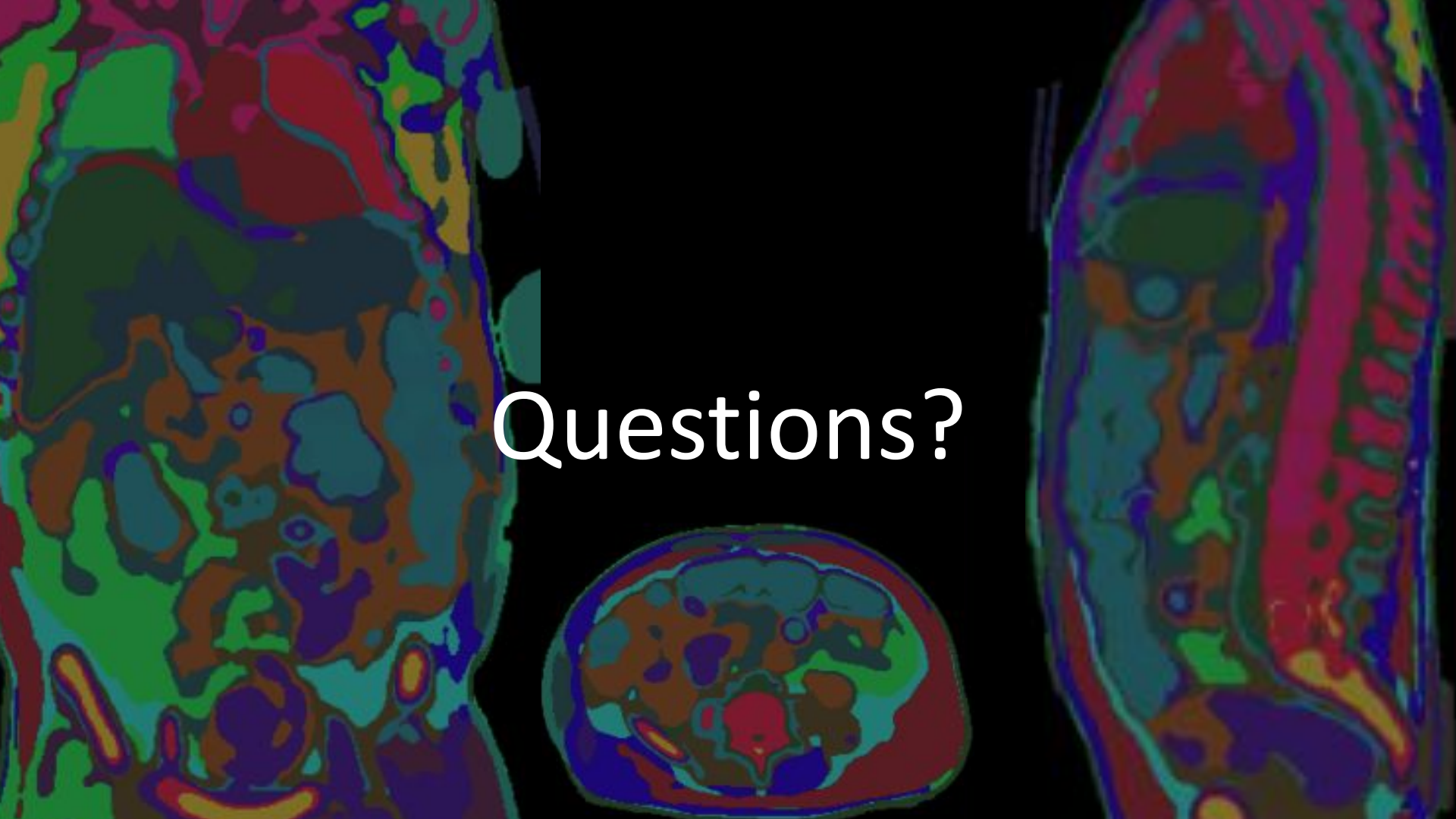
Cochlea L Cochlea R Eustachian tube L Eustachian tube R Internal auditory canal L Internal auditory canal R Mandible L
Mandible R Mastoid L Mastoid R Middle ear L Middle ear R TMJ L TMJ R Tympanic cavity L Tympanic cavity R
Vestibular canal L Vestibular canal R

No analog in supported classes (incl. both GTVs) (14)

Chiasm Eye L Eye R Lens L Lens R Optic nerve L Optic nerve R Parotid L Parotid R Pituitary Submandibular L
Submandibular R GTVnx GTVnd

Plan for Next Week

- Complete SegRap2023 EDA and finalize preprocessing
- Finalize patient-level train/validation/test splits
- Create 5, 10, 20, 40, and 80-patient training subsets
- Set up the nnU-Net baseline training pipeline
- Run initial nnU-Net experiment on the 5-patient subset
- Verify training, inference, and Dice evaluation pipeline

The image features three MRI scans of a human torso, likely the abdominal region, arranged around a central text element. The scans are presented in a false-color format, where different tissues are represented by various colors: reds and oranges for some soft tissues, greens and blues for others, and yellows for specific structures. The top-left scan is a coronal view, showing the internal organs from the front. The top-right scan is a sagittal view, showing the organs from the side. The bottom-center scan is an axial view, showing a cross-section of the torso. The central text 'Questions?' is written in a white, sans-serif font against a black background.

Questions?